

Elementary Functions from a Single Operator

A Summary

Carter Mitchell

Washington State University, USA

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Introduction

Boolean logic and computer circuits can be reduced to the NAND gate. This prompts a similar question of operation reduction for continuous functions and constants on the scientific calculator.

Andrzej Ordzywołek arrived at a binary operator that, while theoretically and practically ineffective, encouraged new questions about pedagogical traditions and the foundations of mathematics.

Exp-Minus-Log

The Single Binary Operator

Ordzywołek found Exp-Minus-Log (EML),

$$\text{eml}(x, y) = \exp(x) - \ln(y),$$

With the constant 1, EML could construct all continuous functions and constants on the scientific calculator.[2]



Depiction of a Realistic Scientific Calculator.[1]

Finding Elementary Functions

The Direct Search Algorithm

Expressing elementary functions $(x, \sqrt{x}, \sin(x), \dots)$ in terms of EML and 1 was difficult. To tackle this, Ordzywołek treated the inputs to EML like a binary tree in which each branch led to different inputs of either EML or 1.

Artificial intelligence models were used to assist in the development of a direct search algorithm that would find a given elementary function.[2]

The Impacts of EML

- Continuous functions as reduced operations.
- Contrived methods for finding expressions.
- Challenged pedagogical tradition for scientific calculators.

References I



FreeSVG.

Scientific calculator.

<https://freesvg.org/scientific-calculator>.

Accessed: September 8, 2026.



Andrzej Odrzywółek.

All elementary functions from a single binary operator.

arXiv preprint arXiv:2603.21852, 2026.